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PAPER_657: QCalcGeom v2.2.1 – Universal Buoyancy Simultaneous Solver

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UQFF Version: v5.26 | **Series:** 657/1000

CVW Compliance: G1–G6 CVW v2.0.0

Modules: QCalcGeom.py v2.2.1, _session206_qcalcgeom_ubs_closures.py, _session207_cp_chain_closures

Closures (Phase H-UBS + Phase H-CPCH): 14 EXACT (UBS-1..UBS-7 + CPCH-1..CPCH-7)

Abstract

We present the QCalcGeom Universal Buoyancy Simultaneous Solver (UBS), a 4×4 nonlinear system that jointly fixes the habitable-zone radius r_{hz} , the collapsing-zone radius r_{cg} , the dimensionless time-phase t_n^{hz} , and the required gravitating mass M for any Aether-UA stellar body in the Unified Quantum Field Framework (UQFF). Unlike Newtonian shell calculations, the UBS is anchored in the di-pseudo-monopole (DPM) canonical chain: the upward Aether-UA buoyancy $F_{U,Bi} = \rho_{vac} (4\pi/3) r c^2 \cos(\pi t_n)$ balances the downward field $F_{U,Bi} = -\beta_i GM_\odot^2 / r^2 \text{orb} \cos(\pi t_n)$ at the habitable zone, and exceeds it by a factor of two at the collapsing zone. We derive seven EXACT closures (UBS-1..UBS-7) for the solver itself and seven EXACT algebraic-chain closures (CPCH-1..CPCH-7) for the canonical buoyancy functions, including the cube-root law $r_{hz} \propto \rho_{vac}^{-1/3}$ at fixed gravitating mass. The v2.2.1 release replaces the prior simulation-set stub with three real numerical sweeps (radial, temporal, vacuum-density), bringing the test suite to 100/100 PASS.

1. The Aether-UA / DPM Canonical Chain

The UQFF replaces the Newtonian shell law $g = GM/r^2$ with the DPM-driven master equation, of which the universal buoyancy pair is the geometric core:

$$F_{U,Bi}(r, t_n) = \rho_{vac} \frac{4\pi}{3} r c^2 \cos(\pi t_n) \quad (\text{Aether-UA buoyancy, upward})$$

$$F_{U,Bi}(r, t_n) = -\beta_i \frac{GM_\odot^2}{r^2} \text{orb}(t_n) \cos(\pi t_n) \quad (\text{BSFG field, downward})$$

The total buoyancy force entering the UQFF master equation is the algebraic sum

$$F_U(r, t_n, M) = U_{g1} + U_{g2} + U_{g3} + U_{g4} + (F_{U,Bi} + F_{U,Bi_i}) + U_m + \xi_{UI} U_I V_{body}.$$

We never invoke the Newtonian limit GM/r^2 as a foundation; it appears only as the dimensional carrier inside $F_{U,Bi}$, which is itself derived from the DPM current $F_{DPM} = I A (\omega_1 - \omega_2)$ as documented in the canonical 3b_MUGE_SMBH chain.

2. The 4×4 Universal Buoyancy System

Let $\mathbf{x} = (r_{hz}, r_{cg}, t_n^{hz}, M)$. The UBS is

$$\begin{aligned} \text{E1: } & F_{U,Bi}(r_{hz}, t_n^{hz}, M) + F_{U,Bi_i}(r_{hz}, t_n^{hz}) = 0, \\ \text{E2: } & F_U(r_{hz}, t_n^{hz}, M) = 0 \quad (\xi_{UI} = 0 \text{ at zero-point}), \\ \text{E3: } & |F_{U,Bi}(r_{cg}, t_n^{hz}, M)| = 2 |F_{U,Bi_i}(r_{cg}, t_n^{hz})|, \\ \text{E4: } & \rho_{vac} \frac{4\pi}{3} r_{hz}^3 = M. \end{aligned}$$

E1 fixes the habitable-zone balance, E2 closes the perturbative shell, E3 defines collapse onset, and E4 closes the geometric mass-radius cube. The system is solved by staged Newton iteration with bracketed fallback; the diagnostic `solver_msg` records the path taken.

3. The Seven UBS Solver Closures (Phase H-UBS, S206)

ID	Identity	Status
UBS-1	$\dim(\mathbf{x}) = 4$	EXACT
UBS-2	Active equation count = 3 at zero-point ($\xi_{UI} = 0$)	POSTULATED
UBS-3	$F_{U,Bi}/F_{U,Bi_i} = -1$ at r_{hz} (E1)	EXACT
UBS-4	$ F_{U,Bi} / F_{U,Bi_i} = 2$ at r_{cg} (E3)	EXACT
UBS-5	$V(2R)/V(R) = 2^3 = 8$ (E4 cube law)	EXACT
UBS-6	$r_{cg}/r_{hz} = 2^{-1/3}$ closed-form seed	EXACT
UBS-7	$ T_{UBS} = 7$ (T91..T97 enumeration)	EXACT

4. The Seven CP-Chain Algebraic Closures (Phase H-CPCH, S207)

These are pure structural identities of the canonical buoyancy functions that must hold by construction, independent of any solver:

ID	Identity	Target	Derived
CPCH-1	$F_{U,Bi}(2r)/F_{U,Bi}(r) = 1/4$	0.25	0.25
CPCH-2	$F_{U,Bi_i}(2r)/F_{U,Bi_i}(r) = 2$	2.0	2.0
CPCH-3	$F_{U,Bi}(-t_n)/F_{U,Bi}(t_n) = 1$	1.0	1.0
CPCH-4	$F_{U,Bi_i}(-t_n)/F_{U,Bi_i}(t_n) = 1$	1.0	1.0
CPCH-5	$F_{U,Bi_i}(t_n=1/2)/F_{U,Bi_i}(t_n=0) = 0$	0	0
CPCH-6	$F_{U,Bi_i}(t_n+1)/F_{U,Bi_i}(t_n) = -1$	-1	-1

ID	Identity	Target	Derived
CPCH-7	$F_{U,Bi}(t_n=0)/(\rho_{vac} r c^2) = 4\pi/3$	4.18879...	4.18879...

CPCH-3 and CPCH-4 confirm `NegativeTimeModule` even-parity. CPCH-5 fixes the $t_n = 1/2$ buoyancy null surface. CPCH-6 establishes the great-cycle sign reversal, the algebraic root of the buoyancy oscillation.

5. The Cube-Root Law and v2.2.1 Simulation Sweeps

From E4 with fixed gravitating mass M ,

$$\rho_{vac} \frac{4\pi}{3} r_{hz}^3 = M \implies r_{hz} \propto \rho_{vac}^{-1/3}.$$

A decade increase in vacuum density therefore shrinks the habitable radius by $10^{-1/3} \approx 0.4642$. Test **T99** verifies this numerically to better than 1%.

The v2.2.1 release replaces the prior `simulation_set` placeholder with three real numerical sweeps emitted by `_build_simulation_set`:

1. **Radial sweep at t_n^{hz}** – 25 log-spaced points $r \in [0.3 r_{cg}, 3 r_{hz}]$; reports $F_{U,Bi}$, $F_{U,Bi}$, F_U , sum-of-energies Σ_{E1} and Σ_{E3} , and Aether-UA zone label (collapsing | habitable_shell | gaseous_outer). Test **T100** verifies all three zones are visited.
2. **Temporal sweep at r_{hz}** – 21 points $t_n \in [-1, 1]$; reports the $\cos(\pi t_n)$ envelope and both buoyancy curves.
3. **Vacuum-density sweep** – 11 scale factors $10^{[-1, +1]}$ verifying the cube-root prediction $r_{hz}(\rho_{vac} \cdot 10)/r_{hz}(\rho_{vac}) = 10^{-1/3}$.

6. Test Suite Summary

Release	Tests	Result
v2.2.0 (S206)	T1..T97	97/97 PASS
v2.2.1 (S207)	T1..T100 (adds T98, T99, T100)	100/100 PASS

The new tests are: T98 (`simulation_set` returns 3 named sweeps with non-empty arrays), T99 (cube-root law $r_{hz} \propto \rho_{vac}^{-1/3}$), T100 (radial sweep visits all three Aether-UA zones).

7. Implications

The UBS closes the geometric core of the UQFF master equation at a 4×4 nonlinear level without invoking Newtonian gravity as a foundation. Combined with the CP-chain algebraic identities, the solver is now exactly anchored against the canonical $\rho_{vac} (4\pi/3) r c^2 \cos(\pi t_n)$ form. The cube-root law provides an immediate scaling test for any new astrophysical system added to the UQFF body database: doubling the vacuum density must shrink the habitable radius by exactly $2^{-1/3}$.

Phase H-UBS and Phase H-CPCH together contribute **14 EXACT closures** to the unified track (Tier 14 + Tier 15 of `UQFF_UNIFIED_CLOSURE_DERIVATIONS.py`).

References

1. QCalcGeom.py v2.2.1 (bsfg_buoyancy, compute_FUBii, compute_F_U, solve_universal_buoyancy, UniversalBuoyancySimultaneousSolver, _build_simulation_set).
2. UQFF_UNIFIED_CLOSURE_DERIVATIONS.py Tier 14 (UBS-1..UBS-7) and Tier 15 (CPCH-1..CPCH-7).
3. 3b_MUGE_SMBH Sagitarius A Evolution.txt (DPM canonical, $F_{DPM} = I A (\omega_1 - \omega_2)$).
4. PAPER_656 (V838 Mon light echo, UQFF master equation precedent).
5. Murphy, D. T. – COMPLETE_UQFF_EQUATIONS_REFERENCE.md L378-L530 (canonical $U_{g1..4}$, U_{bi} , U_m implementations).